Assignment 1 Spring 2025

TTK 4135 Optimization and Control

Adam Godtland Moumen

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Contents

1 Exercise 1 - KKT conditions

1.1 Problem 1: KKT example 1

$$\min x_1 + 2x_2 s.t. 2 - x_1^2 - x_2^2 \geq 0, x_2 \geq 0 \tag{1}$$

1.1.1 1a

Given the constraint $c_1 : 2 - x_1^2 - x_2^2 \geq 0$ the feasible area would be within a circle with radius $\sqrt{2}$.
From the constraint $c_2 : x_2 \geq 0$ the resulting feasible are is within the upper half circle.
The objective function gradient is given by $\nabla f = \begin{bmatrix} 1 & 2 \end{bmatrix}$
Can from these facts conclude that the optimal point would be in $x = (-\sqrt{2}, 0)$. Because from this point there is no direction vector to move along which has a negative dot product with the gradient, and stays in the feasible set Ω.

1.1.2 1b

Lagrangian:

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{i \in \mathcal{E} \cup \mathcal{I}} \lambda_i c_i(x) \tag{2a}$$

$$\mathcal{L}(x, \lambda) = x_1 + 2x_2 - \lambda_1(2 - x_1^2 - x_2^2) - \lambda_2 x_2 \tag{2b}$$

KKT conditions:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0 \tag{3a}$$

$$c_1(x_1^*, x_2^*) \geq 0 \tag{3b}$$

$$c_2(x_1^*, x_2^*) \geq 0 \tag{3c}$$

$$\lambda_i^* \geq 0, \forall i \in \mathcal{I} \tag{3d}$$

$$\lambda_i^* c_i(x^*) = 0, \forall i \in \mathcal{I} \tag{3e}$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = 1 + 2\lambda_1 x_1^* = 0 \tag{4a}$$

$$\lambda_1 = -\frac{1}{2x_1^*} \tag{4b}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 2 + 2\lambda_1 x_2^* - \lambda_2 = 0 \tag{4c}$$

$$\lambda_2 = 2\lambda_1 x_2^* + 2 \tag{4d}$$

$$\text{Inserting the optimal point to find lambda:} \tag{4e}$$

$$\lambda_1^* = -\frac{1}{-2\sqrt{2}} = \frac{1}{2\sqrt{2}} \tag{4f}$$

$$\lambda_2^* = \frac{1}{\sqrt{2}} * 0 + 2 = 2 \tag{4g}$$

$$\lambda_1^* c_1(x^*) = \frac{1}{2\sqrt{2}} * (2-2) = 0 \tag{4h}$$

$$\lambda_2^* c_2(x^*) = 2 * 0 = 0 \tag{4i}$$

As we see,the KKT conditions are satisfied.

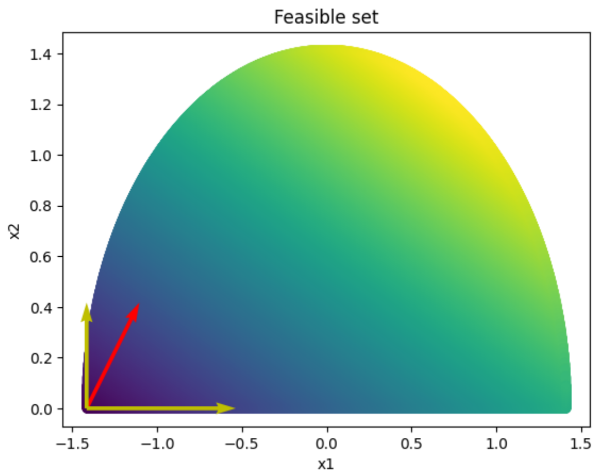
1.1.3 1c

Gradient of constraints and objective function at the solution:

$$\nabla f = \begin{bmatrix} 1 & 2 \end{bmatrix} \tag{5a}$$

$$\nabla c_1 = \begin{bmatrix} 2\sqrt{2} & 0 \end{bmatrix} \tag{5b}$$

$$\nabla c_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} \tag{5c}$$



1.1.4 1d

The lagrange coefficients for inequality constraints need to be positive such that the inequality dosent get flipped by multiplying a negative scalar. By following this, it ensures that lagrangian function is optimal.

1.1.5 1e

The given constraints are convex, because for any set of points in the feasible set, we can draw a line which stays within the set region. The objective function is linear, which means it is also convex. Thus, this optimization problem is convex.

1.2 Problem 2: KKT example 2

1.2.1 1a

$$\min 2x_1 + x_2 s.t. x_1^2 + x_2^2 - 2 = 0 \tag{6a}$$

$$\mathcal{L}(x, \lambda) = 2x_1 + x_2 - \lambda(x_1^2 + x_2^2 - 2) \tag{6b}$$

$$\nabla\mathcal{L} = \begin{bmatrix} 2 - 2\lambda x_1 & 1 - 2\lambda x_2 \end{bmatrix} \tag{6c}$$

$$2 - 2\lambda^* x_1^* = 0 => x_1^* = \frac{1}{\lambda^*} \tag{6d}$$

$$1 - 2\lambda^* x_2^* = 0 => x_2^* = \frac{1}{2\lambda^*} \tag{6e}$$

$$\lambda^* c(x^*) = 0 => \lambda^*(-2 + \frac{1}{(\lambda^*)^2} + \frac{1}{4(\lambda^*)^2}) = 0 \tag{6f}$$

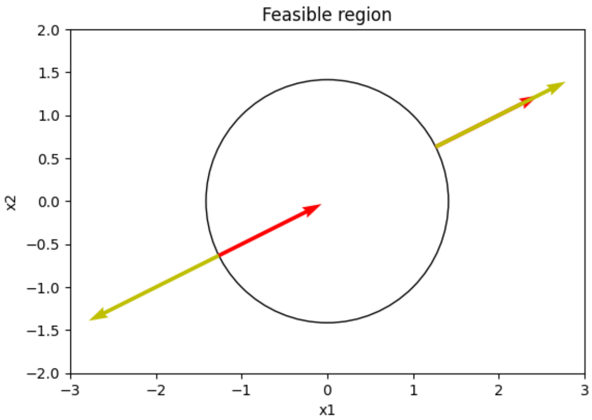
$$-2\lambda + \frac{1}{\lambda} + \frac{1}{4\lambda} = 0 => (\lambda^*)^2 = \frac{5}{8} \tag{6g}$$

$$\lambda^* = \pm\sqrt{\frac{5}{8}} \tag{6h}$$

$$x_1^* = \pm 2\sqrt{\frac{2}{5}} \tag{6i}$$

$$x_2^* = \pm\sqrt{\frac{2}{5}} \tag{6j}$$

1.2.2 2b



1.2.3 2c

The value of the lagrangian multiplier is $\lambda = \pm\sqrt{\frac{5}{8}}$ which makes sense with KKT because the constraint is an equality condition.

1.2.4 2d

Left KKT point is a local solutions because the neighbourhood of points have a bigger f value, while for rightmost point, there is no neighbourhood of points with bigger f value.

1.2.5 2e

$$\omega^T \nabla^2_{xx} \mathcal{L}(x^*, \lambda^*)\omega > 0 \tag{7a}$$

$$\nabla^2_{xx}\mathcal{L} = \begin{bmatrix} -2\lambda & 0 \\ 0 & -2\lambda \end{bmatrix} \tag{7b}$$

Given the point where $\lambda = -\sqrt{(\frac{5}{8})}$, the matrix is positive definite, s.t. the requirment holds. x^* is a strict local solution.

1.2.6 2f

The problem is not convex because the for two points in the feasible set , it cant be drawn a line where every point on the line is in the feasible set. Easier said, the feasible set is along the circlebow.

1.3 Problem 3: Second-order conditions

$$\min -2x_1 + x_2 s.t. \begin{cases} c_1 : (1-x_1)^3 - x_2 \geq 0 \\ c_2 : x_2 + \frac{1}{4}x_1^2 - 1 \geq 0 \end{cases} \tag{8a}$$

The given optimal point is $x^* = (0, 1)$

1.3.1 3a

$$\mathcal{A}(x^*) = 1, 2 \tag{9a}$$
$$-3 * (1-x_1)^2 = -3 \tag{9b}$$
$$\nabla c_1(x^*) = \begin{bmatrix} -3 & -1 \end{bmatrix}^T \tag{9c}$$
$$\nabla c_2(x^*) = \begin{bmatrix} 0 & 1 \end{bmatrix}^T \tag{9d}$$

Because the gradients of the constraints in active set $\mathcal{A}$ are lineary independent, LICQ holds at the optimal point.

1.3.2 3b

$$\mathcal{L}(x, \lambda) = -2x_1 + x_2 - \lambda_1((1-x_1)^3 - x_2) - \lambda_2(x_2 + \frac{1}{4}x_1^2 - 1) \tag{10a}$$
$$\nabla \mathcal{L} = \begin{bmatrix} -2 + 3\lambda_1(1-x_1)^2 - \frac{1}{2}\lambda_2 x_1 & 1 + \lambda_1 - \lambda_2 \end{bmatrix} \tag{10b}$$
$$-2 + 3\lambda_1 = 0 => \lambda_1 = \frac{2}{3} \tag{10c}$$
$$\lambda_2 = 1 + \lambda_1 = \frac{5}{3} \tag{10d}$$

Which are both positive, so KKT conditions are fullfilled.

1.3.3 3c

$$\nabla c_1(x^*) = \begin{bmatrix} -3 & -1 \end{bmatrix}^T \tag{11a}$$

$$\nabla c_2(x^*) = \begin{bmatrix} 0 & 1 \end{bmatrix}^T \tag{11b}$$

$$\mathcal{F} = d^T \nabla c_i(x) \geq 0 \tag{11c}$$

$$\begin{bmatrix} d_1 & d_2 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \end{bmatrix} = -3d_1 - d_2 \geq 0 \tag{11d}$$

$$\begin{bmatrix} d_1 & d_2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = d_2 \geq 0 \tag{11e}$$

$$d_1 \leq -\frac{d_2}{3} \tag{11f}$$

$$\omega_2 = 0 \tag{11g}$$

$$\omega_1 = 0 \tag{11h}$$

$$\omega = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{11i}$$

The critical cone only contains zero vector

1.3.4 3d

Because the 2. order inequality are for vectors excluding zero-vector, and we only have zero-vector in critical cone, we can conclude x^* to be strict local solution.

1.4 Problem 4: Find optimizers

$$\min -x_1 x_2 \tag{12a}$$

$$c_1 : x_1^2 + x_2^2 = 1 \tag{12b}$$

$$\mathcal{L} = -x_1 x_2 - \lambda(1 - x_1^2 - x_2^2) \tag{13a}$$

$$\nabla \mathcal{L} = \begin{bmatrix} -x_2 + 2\lambda x_1 & -x_1 + 2\lambda x_2 \end{bmatrix} \tag{13b}$$

$$\tag{13c}$$